

THE MOUNTAIN PATH ACADEMY

World of Finance

ARIMA Modelling

A Comprehensive Guide to Time Series Forecasting

AR, I, MA Components • Stationarity & ADF Test • ACF/PACF
Box-Jenkins Methodology • SARIMA • Excel Solver • Case Studies

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1 What is ARIMA?

ARIMA — AutoRegressive Integrated Moving Average

ARIMA is one of the most widely used statistical methods for **time series forecasting**. It captures patterns in historical data to predict future values. The model is defined by three parameters: **ARIMA(p, d, q)**.

1.1 The Three Parameters

Param	Name	Meaning	Intuition
p	AR (AutoRegressive)	Number of lagged observations used as predictors	“Today depends on yesterday”
d	I (Integrated)	Times data is differenced to become stationary	“Remove the trend”
q	MA (Moving Average)	Number of lagged forecast errors used as predictors	“Correct recent mistakes”

1.2 The ARIMA Equation

General ARIMA(p,d,q) Model

$$Y'_t = c + \underbrace{\phi_1 Y'_{t-1} + \dots + \phi_p Y'_{t-p}}_{\text{AR part: past values}} + \varepsilon_t + \underbrace{\theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}}_{\text{MA part: past errors}}$$

where Y'_t = differenced series, c = constant, ϕ_i = AR coefficients, θ_j = MA coefficients, ε_t = white noise.

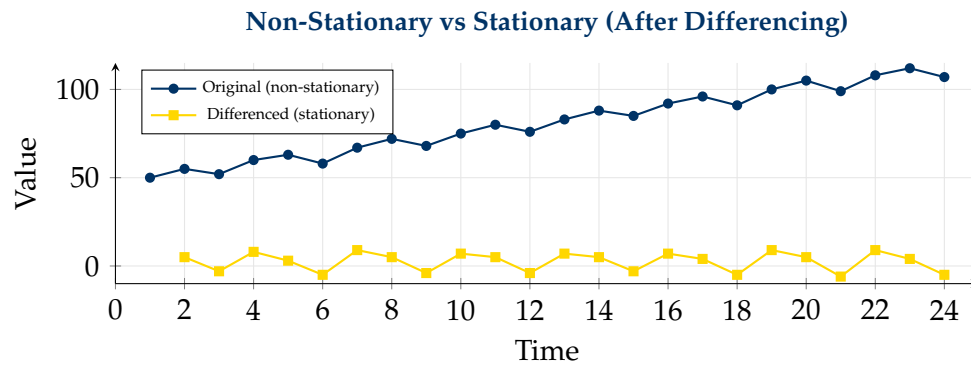
2 Key Concepts

2.1 Stationarity — The Critical Prerequisite

Stationarity

A time series is **stationary** if: (1) constant mean, (2) constant variance, (3) constant autocorrelation structure. ARIMA **requires** stationarity — the d parameter achieves this via differencing.

Test: Augmented Dickey-Fuller (ADF). If p-value < 0.05, the series is stationary.



2.2 Differencing

Differencing — Making Data Stationary

First difference: $\Delta Y_t = Y_t - Y_{t-1}$ (removes linear trend, $d = 1$)

Second difference: $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$ ($d = 2$)

Seasonal difference: $Y_t - Y_{t-s}$ (removes repeating seasonal pattern)

What We Are Trying to Find Out

After differencing, the series should fluctuate around a constant mean. **Over-differencing** introduces artificial negative autocorrelation — if the first ACF lag is strongly negative (near -0.5), reduce d .

2.3 Autocorrelation — ACF and PACF

ACF and PACF

ACF (AutoCorrelation Function): Total correlation between a series and its lags (including indirect effects). Helps determine q (MA order).

PACF (Partial ACF): *Direct* correlation, removing intermediate lag effects. Helps determine p (AR order).

Significance threshold: $\pm 2 / \sqrt{n}$. For $n = 24$, threshold = ± 0.408 .

ACF/PACF Pattern Guide

ACF Pattern	Meaning	Action
Decays slowly (high r at many lags)	Series has a TREND	Apply differencing ($d = 1$ or $d = 2$)
Drops quickly then near-zero	Stationary!	Ready for ARIMA
Spikes at lag $s, 2s, 3s$	SEASONAL pattern	Seasonal differencing ($D = 1$)
Single spike at lag 1	Suggests MA(1)	Set $q = 1$

2.4 Stationarity Check — Worked Example

Lag	ACF (Original)	Stationary?	ACF (Differenced)	Stationary?
1	0.958	No	−0.492	Yes
2	0.953	No	−0.495	Yes
3	0.998	No	0.972	Yes
4	0.946	No	−0.474	Yes

3 Special Cases of ARIMA

Model	Parameters	Description
White Noise	ARIMA(0,0,0)	Purely random; cannot be forecasted
Random Walk	ARIMA(0,1,0)	Best forecast = today's value. Stock prices
AR(1)	ARIMA(1,0,0)	Value depends on immediately preceding value
MA(1)	ARIMA(0,0,1)	Value depends on most recent forecast error
SARIMA	ARIMA(p, d, q)(P, D, Q) _s	Seasonal ARIMA with period s

4 The Box-Jenkins Methodology — 7 Steps

The 7-Step Process

1. **Visualise:** Plot time series. Look for trends, seasonality, outliers.
2. **Test stationarity:** ADF test. If $p > 0.05$, apply differencing.
3. **Determine d :** Difference until stationary. Count = d .
4. **Examine ACF/PACF:** ACF $\rightarrow q$ (MA). PACF $\rightarrow p$ (AR).
5. **Fit candidates:** Try several (p, d, q) . Compare AIC/BIC (lower = better).
6. **Diagnose residuals:** Must be white noise. Ljung-Box test ($p > 0.05$).
7. **Forecast:** Point forecasts + confidence intervals.

If diagnostics fail at Step 6, return to Step 4 with a different model.

AIC and BIC

$$\text{AIC} = 2k - 2\ln(L) \quad \text{BIC} = k\ln(n) - 2\ln(L)$$

k = parameters, n = observations, L = likelihood. Lower = better. BIC prefers parsimony.

5 SARIMA — Seasonal ARIMA

SARIMA Notation

$SARIMA(p,d,q)(P,D,Q)_s$ where P = seasonal AR, D = seasonal differencing, Q = seasonal MA, s = period (4 quarterly, 12 monthly).

Ice Cream Sales — SARIMA(1,1,1)(1,1,0)₄

Seasonal indices: Q1 = 0.468 (winter trough), Q2 = 1.197, Q3 = **1.771** (summer peak), Q4 = 0.564. The model differences both consecutively AND seasonally, capturing the repeating summer/winter pattern.

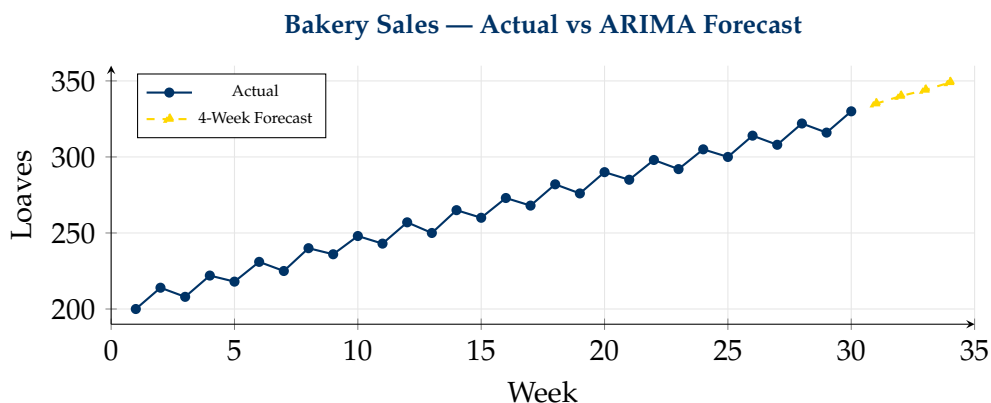
6 Case Study 1: Bakery Bread Sales — ARIMA(1,1,1)

Golden Crust Bakery — 30 Weeks of Data

Parameters: $c = 2.5$, $\phi_1 = 0.40$, $\theta_1 = -0.25$

Equation: $\Delta \hat{Y}_t = 2.5 + 0.40 \times \Delta Y_{t-1} + (-0.25) \times \varepsilon_{t-1}$

"Next week = This week + 2.5 base + 40% of this week's change + error correction"



Accuracy: Mean Residual = 0.50, MAE = 10.81 loaves, RMSE = 10.86 loaves.

7 Case Study 2: Airline Passengers — SARIMA(0,1,1)(0,1,1)₁₂

SkyLine Regional Air — 36 Months, 12-Month Forecast

SARIMA Equation: $\hat{Y}_t = Y_{t-1} + Y_{t-12} - Y_{t-13} + \theta_1 \varepsilon_{t-1} + \Theta_1 \varepsilon_{t-12} + \theta_1 \Theta_1 \varepsilon_{t-13}$

Summer peak (Jul–Aug: 170–190K passengers), winter trough (Nov: 110–120K). Annual growth ≈ 7 –8%.

8 Why ARIMA Fails on Stock Prices

Random Walk = Flat Line Forecast

Stock returns are near-unpredictable. ARIMA converges to ARIMA(0,1,0): $Y_t = Y_{t-1} + \varepsilon_t$. Best forecast = today's price. Returns show near-zero autocorrelation; prices driven by new information, not past patterns (Efficient Market Hypothesis).

9 Excel Solver for Optimal Parameters

Finding Best-Fit Coefficients

1. Enable Solver: File → Options → Add-ins → Solver
2. Set Objective → SSR cell ($\sum \varepsilon_t^2$)
3. Set to: **Min**
4. Changing Variables → c, ϕ_1, θ_1
5. Constraints: $-1 < \phi_1 < 1$ and $-1 < \theta_1 < 1$
6. Method: **GRG Nonlinear**
7. Click **Solve**

Equivalent to Maximum Likelihood Estimation under normal errors.

10 Where ARIMA Works vs Doesn't

Works Well	Why	Example
Retail sales	Strong seasonal + steady trends	Monthly supermarket revenue
Energy demand	Clear seasonal/daily cycles	Daily electricity load
Economic indicators	Gradual momentum	Quarterly GDP, monthly CPI

Fails	Why	Better Alternative
Stock/Crypto prices	Near-random; flat forecast	GARCH, ML models
Pandemic forecasting	Structural break	SIR models, scenario analysis
Multi-factor problems	Ignores external variables	ARIMAX, VAR, ML
Long horizon (> 12)	Converges to mean	Structural models

11 Assumptions and Limitations

6 Key Assumptions

1. **Stationarity** after differencing
2. **Linearity** of relationships
3. **No structural breaks**
4. **Sufficient data** (50+ observations)
5. **No exogenous shocks** (univariate only)

6. Normally distributed errors

Limitations Summary

Limitation	Impact	Mitigation
Univariate only	Misses external drivers	ARIMAX or VAR
Flat long-term forecasts	Reverts to mean	Structural models
Manual tuning	ACF/PACF expertise needed	auto.arima / auto_arma
No regime switching	Can't adapt bull vs bear	Markov-switching models

12 ARIMA vs Other Methods

Method	Strength	Weakness	Use When
ARIMA	Interpretable; less data needed	Univariate; linear only	50–1000 points; stable patterns
Exp. Smoothing	Simpler; intuitive trend/season	Less flexible ACF structure	Quick baseline forecasts
GARCH	Models time-varying volatility	Only variance, not level	Financial risk management
ML (LSTM, XGB)	Nonlinear; many features	Black-box; needs lots of data	> 1000 points; complex data

13 Q&A Practice

Q1: What does ARIMA stand for?

AutoRegressive Integrated Moving Average. AR = regression on own past values. I = differencing for stationarity. MA = modelling past error terms.

Q2: Why does ARIMA fail on stock prices?

Returns are near-unpredictable. ARIMA converges to a random walk: best forecast = today's price. Returns show near-zero autocorrelation. Forecast is a flat line.

Q3: What is the principle of parsimony?

Prefer the simplest adequate model. ARIMA(1,1,1) over ARIMA(3,1,3) if both fit similarly. Simpler models generalise better. BIC enforces parsimony.

Q4: ARIMA vs Machine Learning?

For univariate series with < 1000 points, ARIMA often matches or beats ML. ML excels with many features, large datasets, nonlinear patterns. ARIMA: interpretable, needs less data, easier to validate.

Q5: What is a structural break?

A sudden permanent change (pandemic, regulation). ARIMA estimates from all history equally, so a break contaminates estimates. The model forecasts the *old* regime. Use only post-break data or regime-switching models.

14 Key Takeaways

Chapter Summary

1. ARIMA(p, d, q): p = AR lags, d = differencing, q = MA lags.
2. **Stationarity is mandatory** — test with ADF; fix with differencing.
3. **ACF** determines q ; **PACF** determines p .
4. Follow the **Box-Jenkins** methodology: visualise → test → fit → diagnose → forecast.
5. **SARIMA** extends ARIMA for seasonal data with $(P, D, Q)_s$.
6. Use **AIC/BIC** for model selection; prefer **parsimony**.
7. Residuals must be **white noise** (Ljung-Box test).
8. ARIMA excels for **short-term** (1–12 period) forecasts on stable univariate data.
9. ARIMA **fails** on stocks, structural breaks, and nonlinear data.
10. In Excel, use **Solver** (minimise SSR) for optimal parameters.

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